

LUPIN, NU, Canada

WMO: 714700

Lat: 65.760N

Lon: 111.250W

Elev: 1601

StdP: 13.87

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.9	-36.1	-46.6	0.4	-38.1	-43.3	0.5	-35.2	34.1	-8.1	30.8	-12.4	6.8	270	1.001

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	72.4	57.2	68.8	55.9	64.8	53.9	59.1	69.0	57.0	66.2	54.9	63.3	11.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.8	67.8	62.2	52.6	62.5	60.4	50.5	57.7	58.7	26.5	69.0	25.1	66.3	23.8	63.3	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.0	26.4	23.2	DB	-42.2	78.5	3.4	3.1	-44.6	80.7	-46.6	82.5	-48.5	84.3	-50.9	86.5	(4)
(5)			WB	-41.6	62.1	3.1	1.8	-43.8	63.4	-45.7	64.5	-47.4	65.5	-49.7	66.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	-19.1	-15.1	-9.1	3.5	23.4	44.9	53.4	48.7	37.1	19.0	-1.2	-13.9	(6)
(7)		DBStd	27.71	11.48	10.90	12.64	11.98	10.88	8.88	7.03	7.08	8.13	9.68	10.94	11.18	(7)
(8)		HDD50	13246	2142	1822	1831	1394	824	205	46	110	391	963	1537	1981	(8)
(9)		HDD65	18446	2607	2242	2296	1844	1289	603	361	507	836	1428	1987	2446	(9)
(10)		CDD50	278	0	0	0	0	0	52	152	69	5	0	0	0	(10)
(11)		CDD65	4	0	0	0	0	0	0	3	1	0	0	0	0	(11)
(12)		CDH74	48	0	0	0	0	0	12	26	10	0	0	0	0	(12)
(13)		CDH80	1	0	0	0	0	0	0	0	1	0	0	0	0	(13)
(14)	Wind	WSAvg	10.7	10.1	10.2	10.9	10.5	10.9	10.7	10.3	10.9	12.2	11.9	10.2	9.0	(14)
(15)	Precipitation	PrecAvg	11.5	0.3	0.3	0.6	0.6	0.7	1.0	1.4	2.5	1.7	1.1	0.7	0.6	(15)
(16)		PrecMax	16.6	0.7	0.7	1.5	1.1	1.6	2.5	3.4	6.0	3.3	2.0	1.2	1.2	(16)
(17)		PrecMin	8.8	0.1	0.0	0.2	0.1	0.1	0.0	0.4	1.0	0.3	0.5	0.3	0.1	(17)
(18)		PrecStd	2.0	0.2	0.2	0.4	0.3	0.4	0.7	0.8	1.2	0.8	0.4	0.3	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.4	14.2	23.5	35.4	54.8	75.5	77.3	74.8	61.8	41.7	27.6	17.7	(19)
(20)			MCWB	13.7	13.6	22.6	32.3	45.5	57.0	59.9	57.4	52.0	39.4	27.0	16.8	(20)
(21)		2%	DB	8.8	9.7	19.1	31.5	46.8	69.7	73.5	69.7	56.3	35.8	22.1	13.9	(21)
(22)			MCWB	8.2	8.9	18.3	30.1	40.6	55.2	57.7	57.1	49.2	34.8	21.3	13.3	(22)
(23)		5%	DB	3.2	5.6	15.2	26.6	41.4	64.9	70.2	64.5	51.9	33.3	18.8	8.5	(23)
(24)			MCWB	2.5	5.0	14.4	25.5	37.0	52.9	56.7	54.3	47.4	32.4	18.1	7.8	(24)
(25)		10%	DB	-1.5	1.0	10.1	22.1	37.3	60.5	66.9	60.1	48.5	31.3	14.5	3.1	(25)
(26)			MCWB	-2.2	0.2	9.4	21.1	34.3	50.7	55.2	52.4	45.1	30.6	13.8	2.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	13.6	22.7	32.8	46.1	59.4	62.7	60.4	53.6	39.0	27.1	17.0	(27)
(28)			MDCB	14.3	13.9	23.4	34.9	54.9	72.5	72.3	69.8	59.2	40.0	27.7	17.5	(28)
(29)		2%	WB	8.1	9.1	18.4	29.9	40.7	56.3	60.2	57.6	50.0	34.9	21.5	13.1	(29)
(30)			MDCB	8.7	9.7	19.0	31.5	46.3	67.4	70.3	66.6	54.8	35.5	22.0	13.7	(30)
(31)		5%	WB	2.6	5.0	14.5	25.5	37.2	53.7	58.1	55.6	47.6	32.5	18.0	7.8	(31)
(32)			MDCB	3.1	5.4	15.2	26.6	41.1	64.1	67.3	63.4	51.7	33.1	18.7	8.4	(32)
(33)		10%	WB	-2.1	0.5	9.5	21.2	34.5	51.1	56.2	53.2	45.3	30.4	13.9	2.5	(33)
(34)			MDCB	-1.5	1.1	10.1	22.1	37.2	59.8	64.8	59.0	48.3	31.1	14.5	3.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.7	12.1	14.0	15.6	13.2	16.0	16.8	13.7	10.0	8.4	11.4	11.6	(35)
(36)			MCDBR	14.9	14.8	16.1	15.8	16.5	23.5	21.7	20.6	16.1	8.1	12.9	16.2	(36)
(37)			MCWBR	14.5	14.7	16.0	14.8	11.6	12.4	10.3	10.1	10.1	7.3	13.0	16.1	(37)
(38)		5% WB	MCDBR	14.7	14.5	16.0	15.8	15.7	22.3	20.2	18.8	14.9	7.8	12.9	16.2	(38)
(39)			MCWBR	14.3	14.4	15.9	14.9	11.3	12.3	11.4	10.3	9.9	7.2	13.0	16.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.147	0.222	0.242	0.283	0.311	0.326	0.343	0.309	0.283	0.243	0.153	0.140	(40)	
(41)		taud	1.915	2.147	2.262	2.194	2.263	2.455	2.435	2.537	2.549	2.233	1.915	1.935	(41)	
(42)		Ebn at Noon	118	215	270	281	282	281	272	273	257	202	103	30	(42)	
(43)		Edh at Noon	6	16	24	34	36	30	30	24	19	15	5	1	(43)	
(44)	All-Sky Solar Radiation	RadAvg	35	246	796	1562	2000	1785	1672	1161	621	180	52	7	(44)	
(45)		RadStd	2	13	43	66	146	128	111	104	83	29	10	0	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page